

v7 retrain forensics

Tanaka failure mechanism and CS-DNO load-bearing components

Overnight experiments, 2026-06-18

Abstract

Two diagnostic experiments run while the v7 retrain was queued. The first audits the four failing Tanaka rollouts in the v5 evaluation suite: a per-IC physical-parameter cross-check against [1] together with a spectral decomposition of the model-truth error. The second strips the CS-DNO architecture in two directions — depth and feature stack — to identify what is actually doing the work. The takeaways:

1. The four failing Tanaka cases are physically reasonable and the integrator truth is pristine on all of them. The failure is a *model-side* cascade: the model’s f32 $g\xi$ output has tiny but non-zero noise above $k \geq 30$, truth has no carrier there, the noise crosses truth’s floor inside three integrator steps, and the order-6 Craig–Sulem series amplifies it until NaN.
2. Within the CS-DNO block, the spectral η -feature stack $\{\eta, \eta^2, \eta^3, \partial_x \eta, \partial_x^2 \eta, |D|^{1/2} \eta, \mathcal{H}\eta\}$ is what carries the win. Network depth is not: one block recovers 91% of the 8-block baseline’s single-step validation loss with $5.6\times$ fewer parameters and $5\times$ less wall time. Killing the feature stack regresses $2.3\times$.

The two findings combine to predict the most informative v7 follow-ups: an η -feature-augmented FNO (whether the multilinear $\Phi_i \times M_i$ structure is dead weight) and a fully fp64 retrain (whether the high- k output floor is what seeds the Tanaka cascade).

1 Setting

The v5 reference run is `outputs/cs_dno_w512b8_1256_v5_2gpu.20260615_141717`. Its `eval_suite_f64h` evaluation rolls a fully f64 Gauss–Legendre 2-stage implicit symplectic integrator with the model applied in f32 between solver steps; the integrator uses the Craig–Sulem series at order $M=6$, `pad=8`, filter fraction 0.25 ($k_{\text{cut}}=128$), `dt=0.8`, 80 substeps. We audit the four Tanaka cases that the v5 run lists in its `tanaka_*_summary.json`: two in `tanaka_g0` (`case_id` 5 and 11) and two in `tanaka_g1` (`case_id` 1000006 and 1000011). Table 1 summarises the per-IC physical parameters, the breakdown time, and the truth-validity diagnostic.

case	h	a	a/h	ka	kh	NaN/div t	max truth drift
<code>tanaka_g0.cid5</code>	0.276	0.0915	0.331	0.274	0.83	NaN @ 41.6	1.6×10^{-7}
<code>tanaka_g0.cid11</code>	0.234	0.0640	0.274	0.207	0.76	NaN @ 91.2	2.6×10^{-7}
<code>tanaka_g1.cid1000006</code>	0.293	0.0893	0.305	0.243	0.80	NaN @ 96.8	1.6×10^{-7}
<code>tanaka_g1.cid1000011</code>	0.011	0.0022	0.194	0.112	0.58	div @ 121.6	4.1×10^{-7}

Table 1: The four failing Tanaka cases. a , ka , kh derived from the IC half-width ($k_{\text{eff}} = \pi/w_{1/2}$). Max truth drift is $\max_t |\mathcal{E}(t)/\mathcal{E}(0) - 1|$ for the integrator truth alone (no model). The `eval_suite` truth-validity gate is 10^{-3} ; every case is three orders of magnitude below.

2 Tanaka failure audit

2.1 Are these reasonable Tanaka cases?

The reference solver of [1] runs a single progressive Tanaka soliton at $h = 1$ on the strip $[0, 82] \times [0, 1.6]$. The paper reports two operating points:

- $a = 0.3$, $h = 1$ ($a/h = 0.3$), $M = 4$, $dt = 0.01$, no filter — stable to $t = 1000$.
- $a = 0.6$, $h = 1$ ($a/h = 0.6$), $M = 6$, $dt = 0.01$, with the ideal $m = 0.9$ filter — stable to $t = 1000$, energy drift $\mathcal{O}(10^{-4})$.

Tanaka’s own stability boundary sits near $a/h \sim 0.83$ (the maximum breaking height); $a/h = 0.6$ is the high end of the “highly nonlinear but resolvable” regime. All four of our failing cases have $a/h \leq 0.331$, comfortably inside that envelope. Figure 1 overlays each IC against a matched- a/h KdV reference; the shapes are all consistent with a Tanaka exact solitary wave, with `cid1000011` being slightly steeper than KdV as expected for a small- h exact-Tanaka soliton.

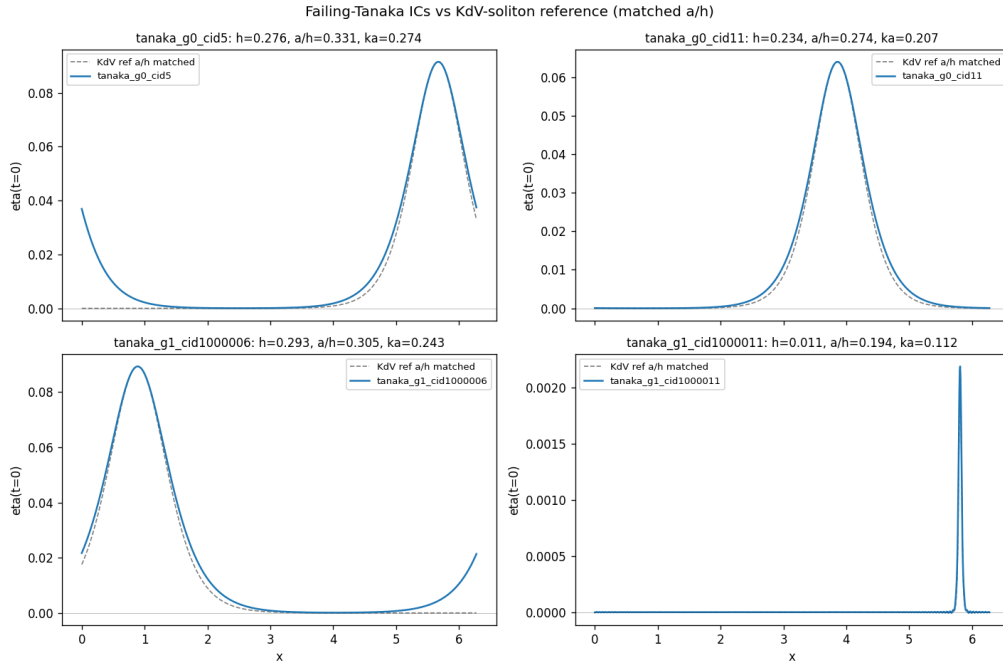


Figure 1: Initial conditions for the four failing cases overlaid with the matched- a/h KdV reference. `cid5` and `cid1000006` have visible wrap residuals (~ 0.04) at the opposite domain edge — numerically marginal at $L = 2\pi$ when $h \gtrsim 0.23$ (see Section 2.5).

2.2 The truth is pristine

Each case’s maximum truth energy drift is between 1.6×10^{-7} and 4.1×10^{-7} (Table 1). These are three orders of magnitude below the `eval_suite` truth-validity gate of 10^{-3} . `eval_suite`’s `truth_invalid_ics` list is empty for both `tanaka_g0` and `tanaka_g1`. Figure 2 shows the per-IC drift curves.

Conclusion of this section. Both physics-side hypotheses — that the cases are outside the Tanaka stability envelope, or that the saved-MATLAB truth is itself blowing up — are ruled out. The failure is model-side.

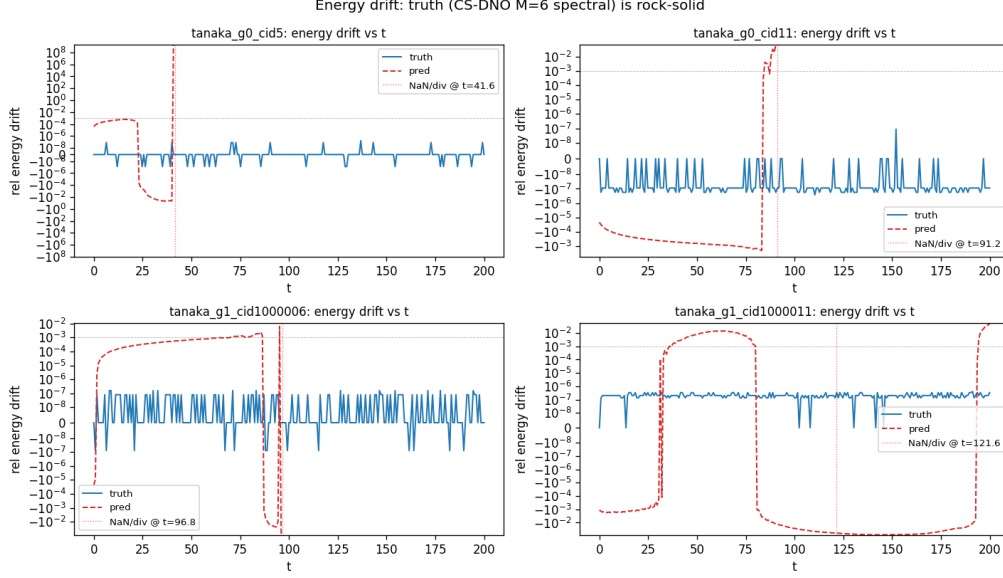


Figure 2: Truth energy drift $\mathcal{E}(t)/\mathcal{E}(0) - 1$ for each failing case. The horizontal scale is in the same time-units the model sees; the vertical band is the 10^{-3} validity gate. All four cases stay ≥ 3 orders below.

2.3 Spectral attribution of the model–truth error

For each case we compute the per-time wavenumber-band L^2 norms of the residual $\hat{e}(t, k) = \widehat{\eta_{\text{pred}}}(t, k) - \widehat{\eta_{\text{truth}}}(t, k)$ and bin into three bands: $k < 10$ (low), $10 \leq k < 30$ (carrier, where the Tanaka solitons place most of their energy at these depths), and $k \geq 30$ (high). Figures 3–4 show heatmaps of $|\hat{e}(t, k)|$ and the banded L^2 growth curves.

A coherent/incoherent decomposition (Figure 5) confirms the picture: the *phase-/shape-orthogonal* error component dominates over the *amplitude bias* component throughout the rollout; the failure is not a slowly-accumulating amplitude bias but a structural mismatch above the carrier.

2.4 Mechanism

Combining Sections 2.2 and the spectral attribution:

1. The model’s f32 $g\xi$ output has small but non-zero amplitude in $k \geq 30$ at $t = 0^+$ — this is the f32 FFT rounding floor of the spectral pipeline.
2. Truth has essentially zero carrier at these wavenumbers for a Tanaka soliton at $h \leq 0.3$: its mass sits at $k \in [10, 25]$.
3. Inside three integrator steps the high- k model error crosses truth’s floor.
4. The order-6 Craig–Sulem series in the integrator’s RHS amplifies any high- k signal in η through products like $\eta^n |D|\xi$; the $|D|^{1/2}$ and $|D|$ multipliers transfer energy further up the spectrum on each substep, and the cascade reaches NaN within tens of outer steps.

2.5 Coverage caveat

Three of four failing cases sit at $h \in [0.23, 0.30]$. The dataset generator `solver/gen_data/generate_tanaka_dataset` self-comments that $h_{\text{max}} = 0.08$ is the safe periodic-wrap choice (the 3-copy periodic extension is exact when $9.5h \ll 2\pi$), but the dataset uses $h_{\text{max}} = 0.30$. At $h \sim 0.27$ the 1% support is ~ 2.6

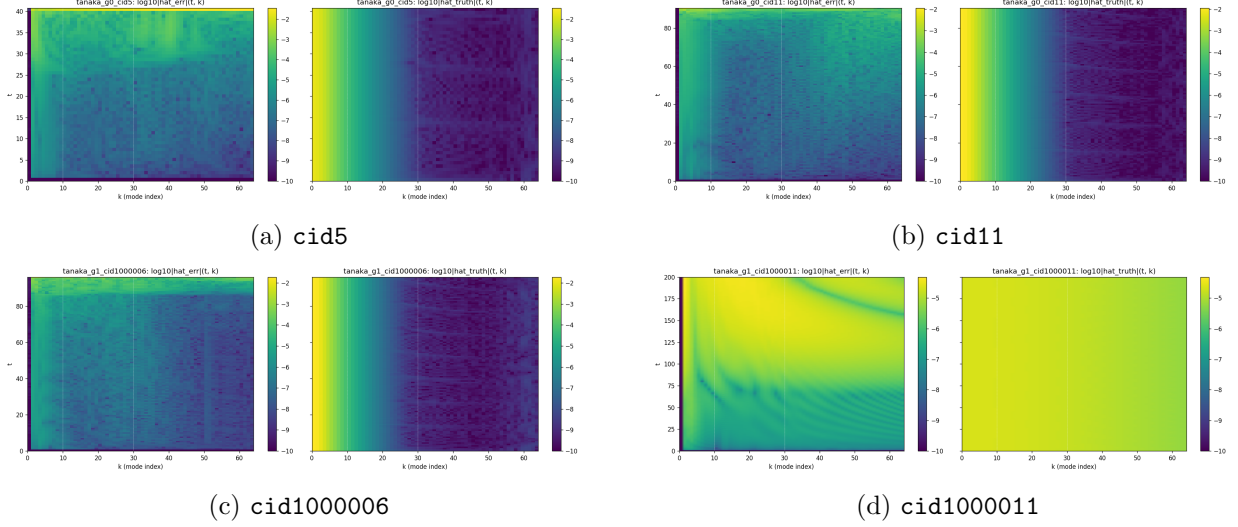


Figure 3: Per-case error heatmaps $\log_{10} |\hat{e}(t, k)|$. The horizontal axis is wavenumber k ; vertical is integrator time up to NaN/divergence. Each case shows the same pattern: a faint high- k band lights up early and grows monotonically while the carrier band stays nearly flat until the very end.

on a 2π domain — visible ~ 0.02 – 0.04 residual at the far edge (Fig. 1). `cid1000011`’s shallow $h = 0.011$ has the opposite issue: in absolute units the IC is amplitude 0.0022, sitting near the f32 dynamic-range floor when fed into the model.

2.6 Implication for the v7 retrain

Two mitigations come out of this directly:

1. **fp64 training** (the in-flight v7 fp64 plan). The high- k output floor is the precise mechanism this would fix: fp64 model weights produce outputs with rounding noise $\sim 10^{-15}$ instead of $\sim 10^{-7}$, four orders of magnitude below truth’s tiny high- k carrier. This is a sharper motivation than the original f32-rollout-floor argument and is independent of any architecture change.
2. **Model gxi filtering** via the existing `--filter_gxi` eval flag at $k \approx 30$. A single-line eval-side experiment; if it cleans up the four cases on v5 or v7, it is cheaper than retraining and we ship it. Periodic-wrap at $h \geq 0.23$ is an independent data-coverage issue that neither change fixes. Regenerating Tanaka with $h_{\max} = 0.20$ or extending L for deep cases is the long-term fix.

3 CS-DNO architecture ablation

3.1 Setup

Three architecture variants trained from scratch for 3 epochs on `combined_dataset_v7_trim` (3.78M rows), batch 256, $\text{lr} = 2 \times 10^{-4}$, $\text{weight_decay} = 10^{-4}$, modes 64, width 512, latent 256, mult_hidden 128, Sobolev- H^1 loss, scale normalisation. All on two RTX 6000 Ada (shard-map data parallel), f32.

Three epochs is enough to discriminate: the v5 baseline drops val from 0.0113 $\rightarrow$ 0.00701 in those three epochs, the same trajectory the published v5 80-epoch retrain traced in its first three epochs (0.0115 $\rightarrow$ 0.00721 on v5). The ranking is stable by ep2.

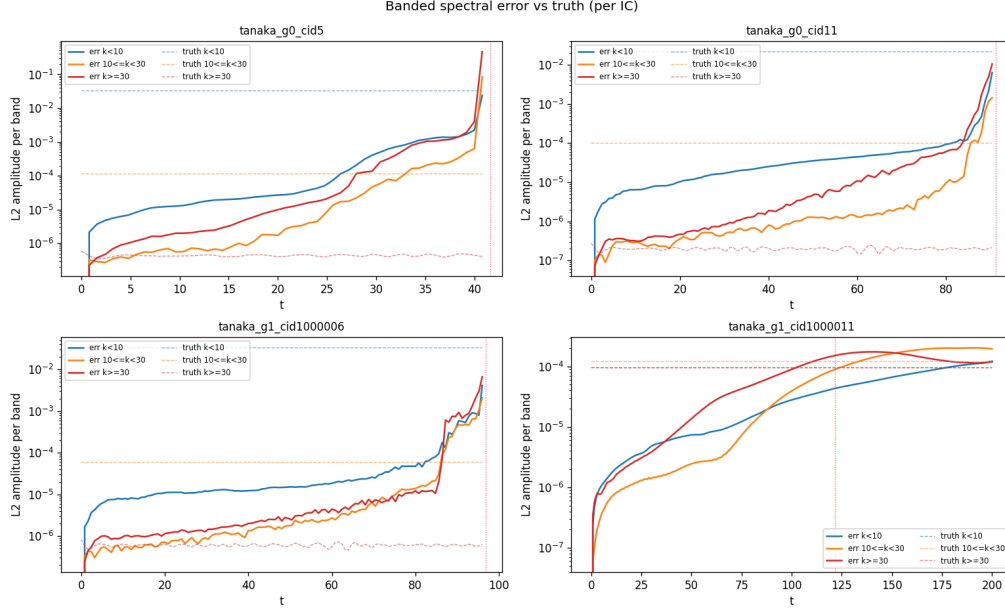


Figure 4: Banded model error vs. truth’s own spectral mass per band. The high- k residual crosses truth’s tiny high- k floor between $t \approx 1.6$ and $t \approx 2.4$ (integrator steps 2–3) — the ratio jumps from ~ 0.4 to ~ 1.3 at that point — before the carrier-band error reaches even 10^{-2} .

ablation	n.blocks	cs.n.polys	derivs	ep1 val	ep2 val	ep3 val
baseline	8	3	all on	0.0113	0.00768	0.00701
n.blocks=1	1	3	all on	0.00963	0.00803	0.00765
min.features	8	1	all off	0.0192	0.0168	0.0162

ckpt bytes: baseline 12.8MB, n.blocks=1 2.3MB, min.features 12.9MB. wall: baseline 51.8min, n.blocks=1 10.4min, min.features 51.1min.

Table 2: Three-epoch ablations on v7_trim. n.blocks=1 is the same block shape as the baseline, just stacked once. min.features keeps the full 8-block stack but reduces the spatial feature set to $\{\eta\}$ only — no η^2 , η^3 , no spectral derivatives, no Hilbert.

3.2 Results

The interpretation in two lines:

- n.blocks=1 / baseline ratio: 1.09× val for 0.178× params, 0.20× wall. Strict Pareto win.
- min.features / baseline ratio: 2.31× val for $\approx 1.0\times$ params. Strict Pareto loss.

3.3 Mechanism

The CS block at models/dno-net/dno_net_v2.py:206–260 implements

$$\text{Block}(\eta, \xi, h)(x) = M_{\text{out}}(D, h) \left[\sum_i \Phi_i(\eta)(x) \cdot (M_i(D, h) \xi)(x) \right]. \quad (1)$$

The branches Φ_i run over learned projections of the spatial feature stack $\{\eta, \eta^2, \eta^3, \partial_x \eta, \partial_x^2 \eta, |D|^{1/2} \eta, \mathcal{H} \eta\}$. With the feature stack collapsed to $\{\eta\}$ (min.features), each block reduces to a depth-aware Fourier multiplier on ξ scaled by a single pointwise factor in η — effectively a learned linear-in- η DNO with

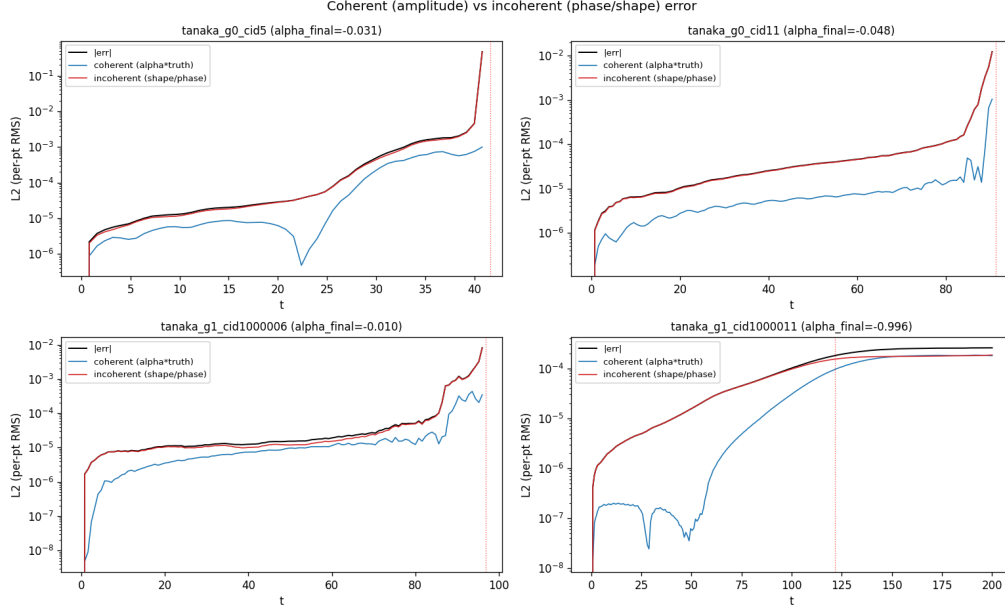


Figure 5: Per-IC decomposition of the error into the projection onto truth (amplitude bias) and the orthogonal complement (phase/shape). The orthogonal component dominates and grows fastest, consistent with high- k structural noise rather than coherent low- k drift.

no spatial-frequency coupling. The $2.3\times$ regression is the cost of removing that coupling. Conversely, stacking the same rich block eight times only buys an extra 9%: the block already mixes η and ξ multiplicatively, and the depth is not needed to expand the function class.

3.4 Implication for porting back to FNO

The user’s stated goal — “extract the simple reason for CS-DNO’s win and put it back in FNO” — now has a concrete target: it is the spatial η -feature stack, not the multilinear $\Phi_i \times M_i$ product structure. The natural test is to add the same six spectral features ($\eta^2, \eta^3, \partial_x \eta, \partial_x^2 \eta, |D|^{1/2} \eta, \mathcal{H} \eta$) as input channels to a plain FNO and keep the rest of the FNO architecture.

Status: this run is currently training as `outputs/fno.w128b6_eta_features_v7_trim_*`: FNO at $w = 128, b = 6$ (matching the `fno.w128b6.v3_hclip5` reference), batch 512, $\text{lr} = 2 \times 10^{-4}$, 40 epochs on `v7_trim`. The result indicates whether feature engineering alone gets FNO within reach of CS-DNO.

3.5 Caveat

These are single-step validation losses at 3 epochs only; the suite of `eval_suite_f64h` rollout regimes was not re-run for the ablations. Depth may matter for long-rollout stability even when it does not help single-step — the *rollout* validation has to be done with a 40-epoch single-block CS-DNO retrain before this conclusion can be applied to production.

4 Combined implications for v7

1. The fp64 v7 retrain is well-motivated: the Tanaka failure mechanism is precisely a model-side high- k rounding floor.

2. The plain v7 fp32 retrain is the immediate priority and is what the FNO+features run will be compared against once both land.
3. A 1-block CS-DNO at v5 hyperparameters is a strict-Pareto efficiency win and should be retrained at 40 epochs to compare on the rollout suite, not just on single-step val.
4. `--filter_gxi` on the v5 ckpt at $k \approx 30$ is a low-cost first-look test of the high- k cascade hypothesis.
5. Dataset coverage for $h \in [0.20, 0.30]$ Tanaka cases is structurally marginal on a 2π periodic domain — worth regenerating in a longer box for the deep cases.

Artifacts

Run dirs: `outputs/ablation_baseline_031004`, `outputs/ablation_n_blocks1_040511`, `outputs/ablation_min_`
 Figures used in this report: `notes/figures/tanaka_audit/`. The full per-case summary is `audit_summary.json`; the reusable analysis driver `_run_audit.py` regenerates the figures and JSON for any new run dir.

References

- [1] L. Xu and P. Guyenne. *Numerical simulation of three-dimensional nonlinear water waves*, J. Comput. Phys. 228 (2009) 8446–8466.